Hugo Marquerie 14/06/2026

Teo Fn Holomorfa Disco Unidad No Anula Cota Derivada

Teorema 1. *Sea $f \in \mathcal{H}(\mathbb{D})$ tal que $f(\mathbb{D}) \subset \mathbb{D} \setminus \{0\}$. Entonces, $\forall z \in \mathbb{D}$:*

$$\left(1 - |z|^2\right) |f'(z)| \leqslant 2\, |f(z)| \, \log\left(\frac{1}{|f(z)|}\right).$$

En particular, $|f'(0)| \leqslant 2\, |f(0)| \, \log\left(\frac{1}{|f(0)|}\right)$.

***Demostración*:** Como $\mathbb{D}$ es simplemente conexo y f no se anula en $\mathbb{D}$, por el `teo-fn-holomorfa-dominio` existe $g \in \mathcal{H}(\mathbb{D})$ tal que $e^g = f$. Entonces, como $f(\mathbb{D}) \subset \mathbb{D}$, $|f| = e^{\Re(g)} < 1$, luego $\Re(g) < 0$. Por tanto, $(-g) \in \mathcal{H}(\mathbb{D})$ y $(-g)(\mathbb{D}) \subset \mathbb{H}$.

Entonces, por la `prop-subordinacion-parte-real-positiva/`**??**, $\forall z \in \mathbb{D} : |(-g)'(z)| \left(1 - |z|^2\right) \leqslant 2\Re(-g(z))$.

- Para el lado izquierdo, como $f'(z) = e^{g(z)} g'(z) = f(z) g'(z)$, obtenemos que $|(-g)'(z)| = |g'(z)| = \frac{|f'(z)|}{|f(z)|}$.
- Para el lado derecho, tenemos que $\Re(-g(z)) = -\Re(g(z)) = -\log|f(z)| = \log\left(\frac{1}{|f(z)|}\right)$.

Por tanto, $\forall z \in \mathbb{D} : \left(1 - |z|^2\right) |f'(z)| \leqslant 2\, |f(z)| \log\left(\frac{1}{|f(z)|}\right)$.

En particular, para $z = 0$ obtenemos que $|f'(0)| \leqslant 2\, |f(0)| \log\left(\frac{1}{|f(0)|}\right)$. ■

Referenciado en

- Teorema del cubrimiento de Landau

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.